Character sets: ASCII and Unicode

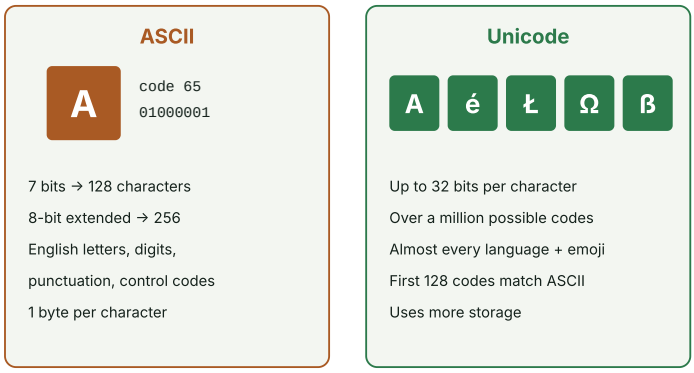
D06 FACT FILE

Specification: students should be able to

- Demonstrate understanding of the following types of character representation: ASCII (7-bit and 8-bit) and Unicode

CCEA fact file: Digital Data

Key words: character set, ASCII, extended ASCII, Unicode, character code, control character, emoji



ASCII compared with Unicode

ASCII

The American Standard Code for Information Interchange originally used 7 bits, giving 128 codes: 32 control codes followed by the space, digits, punctuation and upper and lower case English letters. Extended ASCII uses 8 bits for 256 characters, adding some accented letters and symbols. ASCII is efficient but cannot represent most of the world's languages.

Unicode

Unicode is an international standard that aims to give every character in every writing system its own code, including emoji. It uses more bits per character than ASCII, so text can take more storage, but it allows documents to be shared worldwide. Its first 128 codes match ASCII.

T Key terms from the mark schemes

ASCII
American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English.
Credited: U4 2019 Q7, 2023 Q1, 2025 Q4

Unicode
Character set covering almost all languages and emoji, using up to 32 bits per character; includes ASCII.
Credited: U4 2019 Q7, 2023 Q1, 2025 Q4

Exam focus

Questions include expanding ASCII (2019 Q7(a), 2023 Q1(a)), the capacity of 7-bit ASCII (2023 Q1(b)), true/false statements (2019 Q7(b), 2023 Q1(d)), and advantages/drawbacks of each (2025 Q4(d)).

Unit 4 2025 Q4	[15]	Binary conversion, addition and overflow, units, Unicode vs ASCII
Unit 4 2023 Q1	[10]	ASCII acronym and capacity; AND/OR truth table; Unicode vs ASCII
Unit 4 2022 Q1(d)–(g)	[4]	Abstraction, decomposition, search efficiency, ASCII
Unit 4 2019 Q7	[5]	ASCII and Unicode

? Check yourself

1. Expand the acronym ASCII. [1]
2. How many characters can be represented in 7-bit ASCII? Explain. [2]
3. The ASCII code for 'A' is 65. What is the code for 'E'? Give it in denary and binary. [2]

Answers: 1. American Standard Code for Information Interchange (1) 2. 128 characters (1) / because 7 bits give 2^7 = 128 different binary patterns (codes 0–127) (1) 3. E is 4 letters after A, so 65 + 4 = 69 (1) / 01000101 (1)